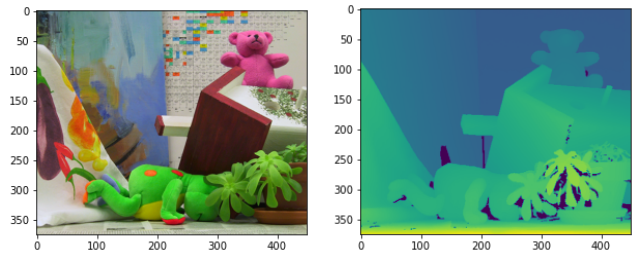
CS 4670/5670: Computer Vision, Spring 2026

Project 3: Stereo



Brief

- Released: Monday, March 16th, 2026
- Due: Wednesday, April 8th, 2026 at 11:59PM (turn in via CMSX)
- Teams: This assignment should be done in a group of at most 2 students.

Synopsis

This assignment is meant to exercise concepts of stereo and photometric stereo. As a recap, the goal of stereo is to reconstruct the 3D positions of points on objects using 2 cameras with known intrinsic and extrinsic parameters, and the goal of photometric stereo is to estimate the surface normals of an object using a fixed camera and multiple lighting positions. There are two parts to this assignment, please complete the specified functions in the following files.

- `stereo.py`
- `photometric_stereo.py`

Getting Started

Python Packages

Please use python3 for this assignment. We recommend creating a virtual environment and installing all the necessary packages via this command to limit issues with setup.

```
pip install numpy matplotlib Pillow scipy
```

Dataset

Download, unzip, and place the [data folder] in the same directory as your code.

Part 1 - Stereo:

For this part of the assignment, there are 3 functions to complete: `get_ncc_descriptors`, `compute_ncc_vol`, and `get_disparity`.

Please refer to `stereo.py` for instructions.

Part 2 - Photometric Stereo:

For this part of the assignment, there is 1 function to complete: `photometric_stereo`.

Please refer to `photometric_stereo.py` for instructions.

Demo:

There are two demo notebooks that you can use to test your implementation of stereo and photometric stereo. These are called `stereo_demo.ipynb` and `photometric_stereo_demo.ipynb` respectively.

We have also provided a version of our complete demo notebooks as html files which can be used to compare your test case outputs with ours. The html files are called `stereo_demo.html` and `photometric_stereo_demo.html`.

Turn in:

To recap, you must submit:

- `stereo.py`
- `photometric_stereo.py`